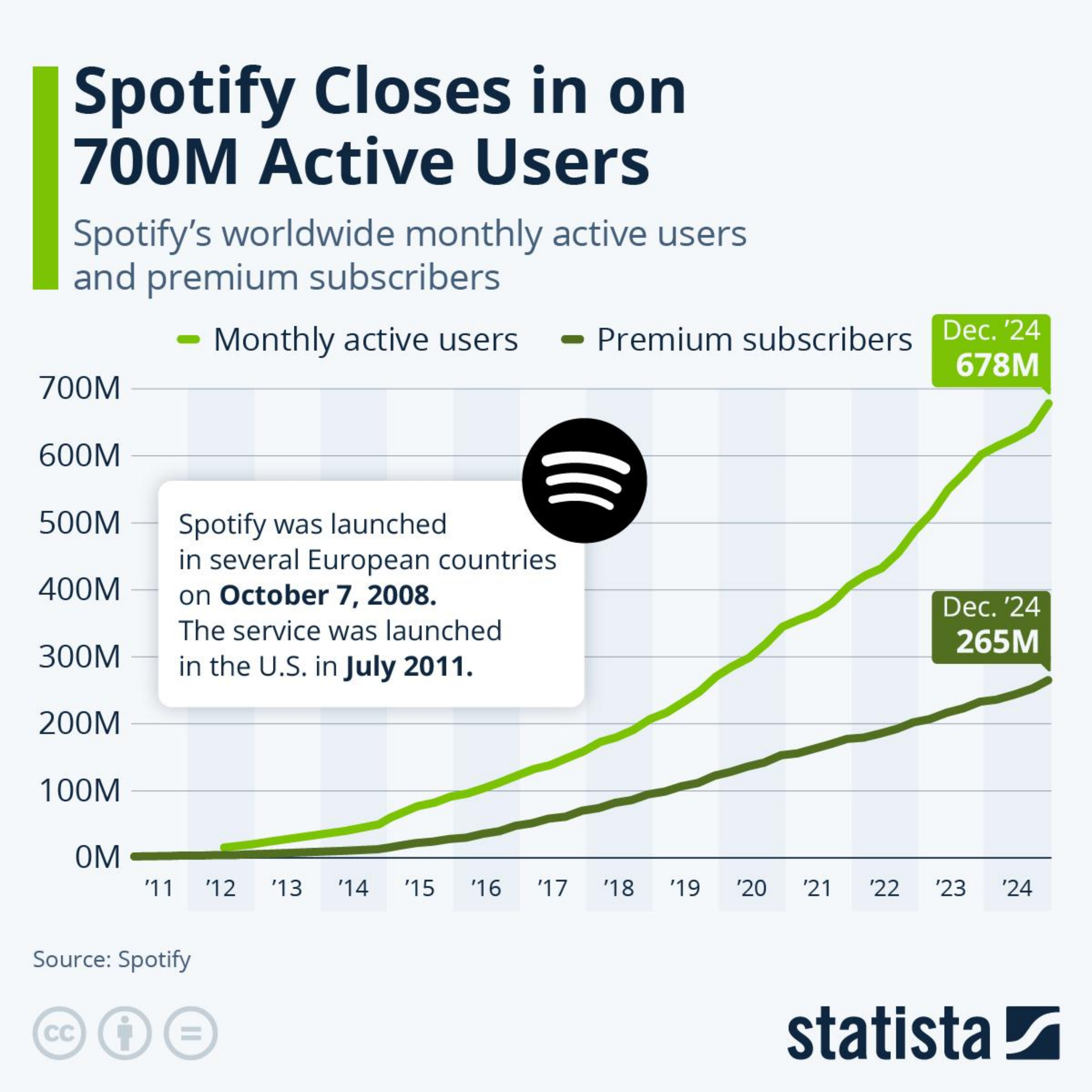
# Science of Song Success

STA 160 Capstone Project | University of California Davis

Capri, Rohan, Alex, Nick, Edward

# Why Study Song Popularity?

- Streaming platforms, like Spotify and Youtube, generate millions of data points per song.
- Record labels rely on this feedback to decide which artists and genres to promote.
- Our Goal: Use machine learning to understand what drives success across Spotify and Youtube.



# Our Research Question

“Which song-level features best predict popularity, and how can predictive models guide record labels in genre and marketing decisions?”

Objectives:

- Identify key predictors of success
- Build interpretable and accurate predictive models
- Create a dashboard to visualize results

# Spotify-Youtube Dataset

- Source: Kaggle (Rohit Grewal, 5,700+ songs)
- Combines Spotify audio features (danceability, energy, etc.) and Youtube engagement data (views, likes, comments, etc.).

*Snapshot of artist and url spotify values via Kaggle*

**Artist**

The person or band who performed the song

**2079**
**unique values**

| | | |
|---|---|---|
| Valid | 20.7k | 100% |
| Mismatched | 0 | 0% |
| Missing | 0 | 0% |
| Unique | 2079 | |
| Most Common | Gorillaz | 0% |

**Url_spotify**

Spotify Link

**2079**
**unique values**

| | | |
|---|---|---|
| Valid | 20.7k | 100% |
| Mismatched | 0 | 0% |
| Missing | 0 | 0% |
| Unique | 2079 | |
| Most Common | https://ope... | 0% |

# Data Preparation and Ethics

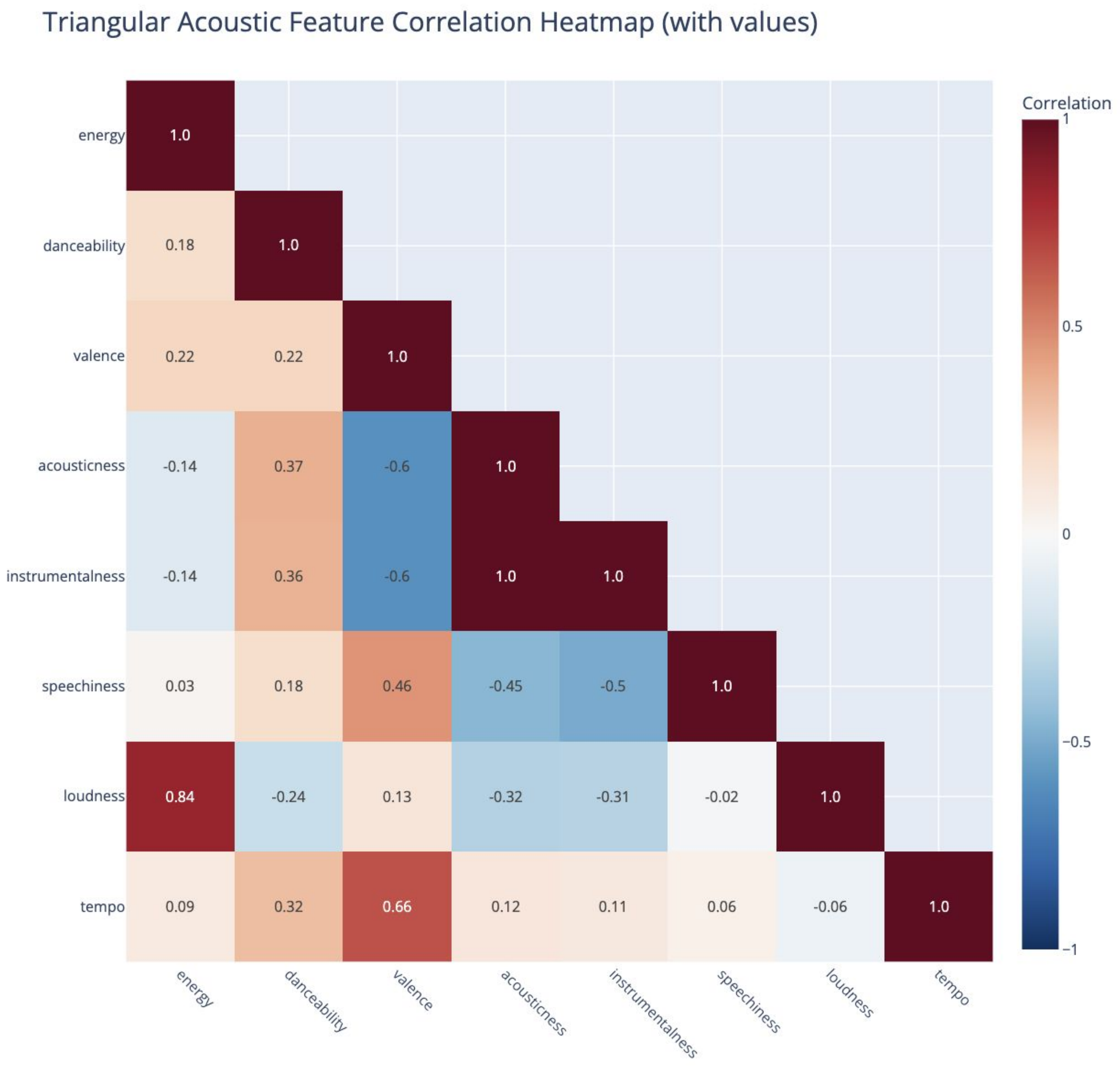
- Removed duplicate and missing values
- Normalized engagement metrics all platforms
- Verified variable ranges (0-1 for audio features)
- Data is public, aggregated, and ethically sourced

Raw Data → Cleaning → Normalization → Validation

# Initial Insights

- Energy and danceability positively correlated with views and streams.
- Engagement metrics show heavy right skew, meaning few viral/extremely popular hits.
- Genres differ in average popularity levels.

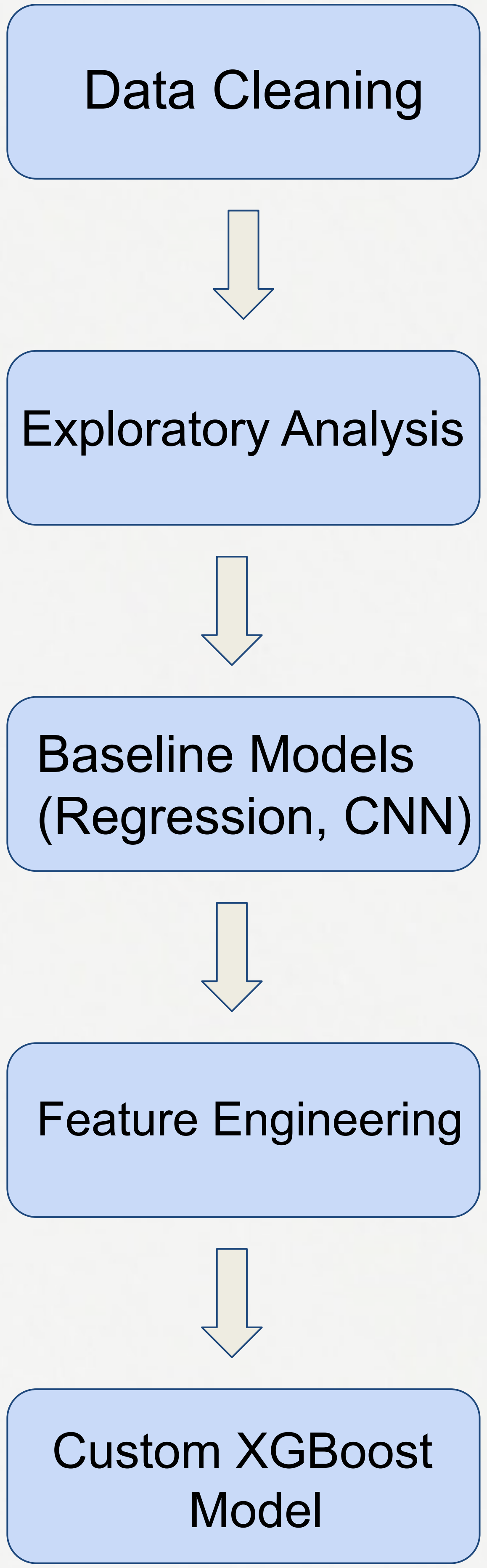
*Snapshot of Correlation Heatmap via Website*




*This triangular acoustic heatmap displays correlations among audio features such as energy, danceability, valence, and tempo.*

# Modeling Workflow

- Started with regression models for interpretability
- Tested Random Forest and early XGBoost (low accuracy)
- Built our own XGBoost-based model after transfer learning failed



# Building Our Own Model

- Added feature interactions (energy x danceability, tempo x valence).
- Engineered audio embeddings for rhythm and timbre.
- Split data (80% training, 20% validation).
- Tuned learning rate, depth, and regularization.

- Regression
Interpretable baseline

→

- Random Forest/CNN
Severe Underfitting

→

- Basic XGBoost
Weak Features

→

- Feature Engineering
Interactions + Embeddings

→

- Custom XGBoost
$R^2 = 0.70$

# Model Results

- Marketability model outperformed Popularity model
- Both fall within the range of published studies ($R^2$: 0.45–0.75).

*Table 1: Model Performance Summary*

| Model | Coefficient of Determination ($R^2$) | Mean Absolute Error |
|---|---|---|
| Popularity | 0.656 | 5.021 |
| Marketability | 0.698 | 2.017 |

# Hybrid Similarity System

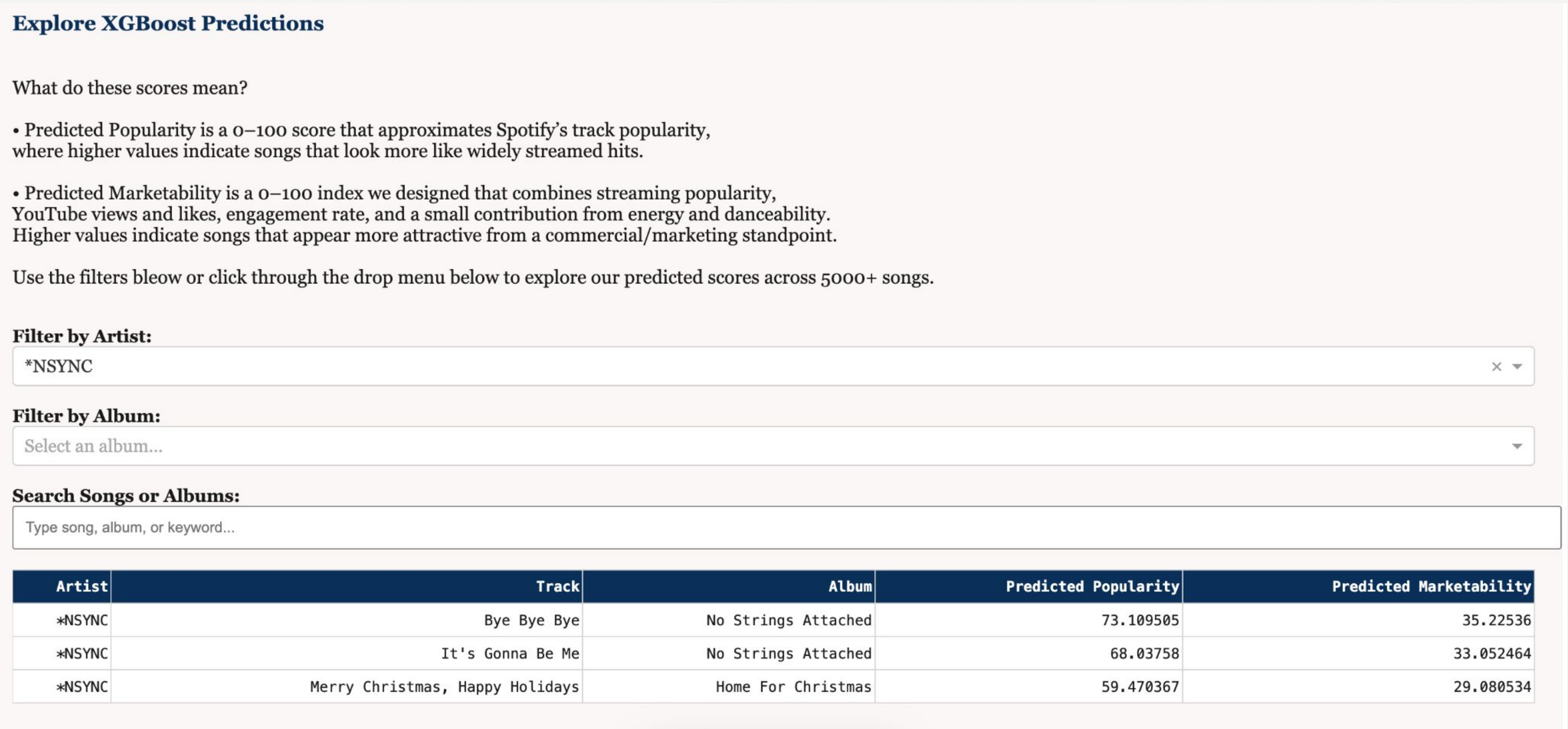
What it does

- Identifies the top 5 most similar songs to a given track
- Uses both predicted popularity and predicted marketability
- Accounts for audio features and listener engagement

Why it matters

- Demonstrates recommendation system potential
- Show how music and listener behavior work together

*Snapshot showing predicted popularity/marketability via website*



**Explore XGBoost Predictions**

What do these scores mean?

• Predicted Popularity is a 0–100 score that approximates Spotify's track popularity, where higher values indicate songs that look more like widely streamed hits.

• Predicted Marketability is a 0–100 index we designed that combines streaming popularity, YouTube views and likes, engagement rate, and a small contribution from energy and danceability. Higher values indicate songs that appear more attractive from a commercial/marketing standpoint.

Use the filters bleow or click through the drop menu below to explore our predicted scores across 5000+ songs.

**Filter by Artist:**

*NSYNC

**Filter by Album:**

Select an album...

**Search Songs or Albums:**

Type song, album, or keyword...

| Artist | Track | Album | Predicted Popularity | Predicted Marketability |
|---|---|---|---|---|
| *NSYNC | Bye Bye Bye | No Strings Attached | 73.109505 | 35.22536 |
| *NSYNC | It's Gonna Be Me | No Strings Attached | 68.03758 | 33.052464 |
| *NSYNC | Merry Christmas, Happy Holidays | Home For Christmas | 59.470367 | 29.080534 |

# What Drives Song Success?

Audio Features
- Energy
- Danceability
- Tempo
- Valence

**Popularity = Music * Listener Behavior**

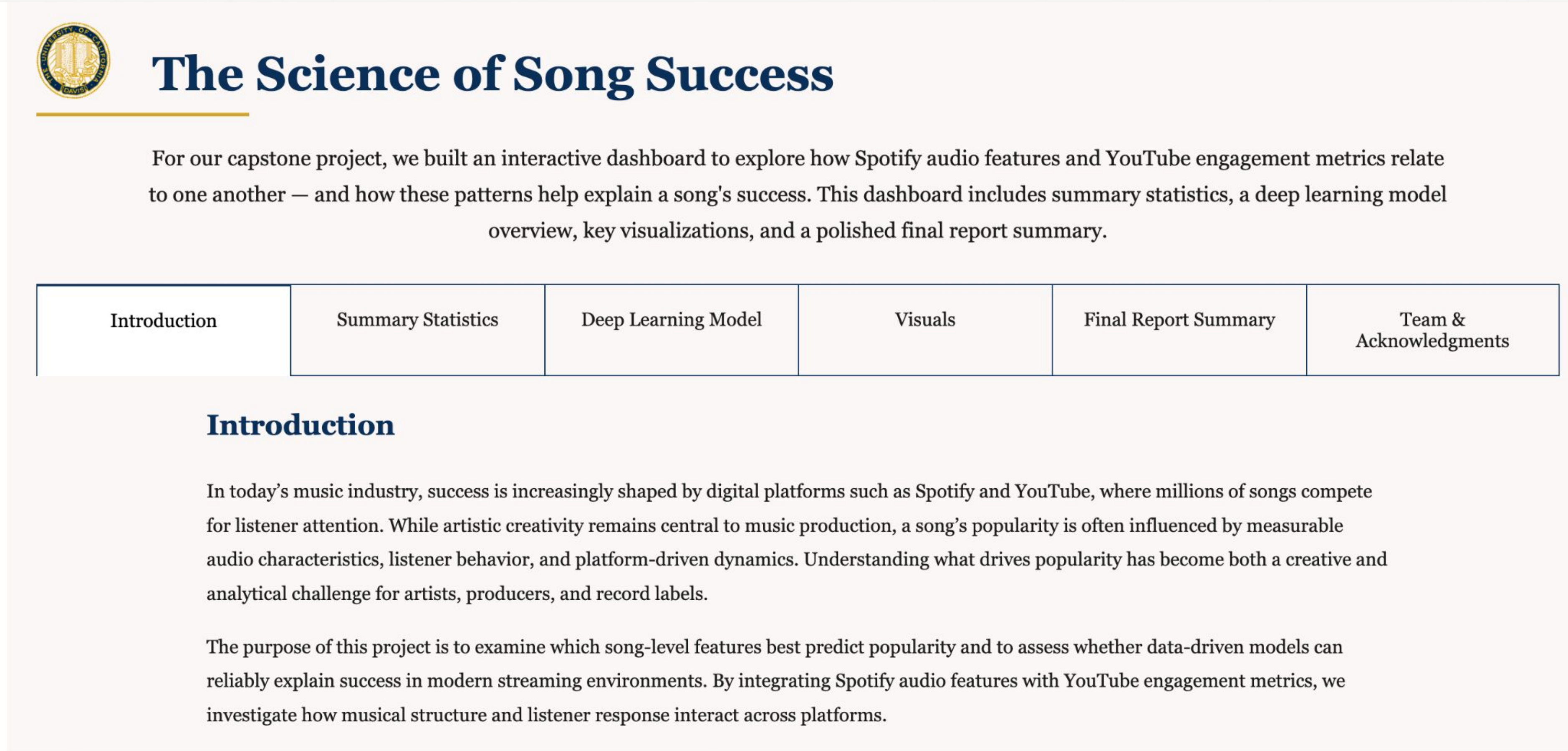
Engagement Metrics
- Views
- Likes
- Comments

Engagement metrics had stronger predictive power, but audio features improved accuracy.

# Interactive Dashboard

- Built with Dash and Plotly.
- Tabs for summary statistics, model results, and interactive filters.
- Allows record labels and artists to explore relationships visually.

*Snapshot of first page of website*



# Conclusion

Key Takeaways

- Built a custom machine learning model when pre-trained models failed

- Achieved strong predictive performance ($R^2 \approx 0.65$–$0.70$)

- Found that listener engagement drives popularity more than audio alone

- Demonstrated potential for music recommendation and industry use

# Acknowledgments


- STA 160 Faculty and TAs
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- UC Davis Statistical Data Science Department


# Thank you!